

GENETIC, EVOLUTIONARY, AND BIOGEOGRAPHICAL PROCESSES IN INTERMITTENT RIVERS AND EPHEMERAL STREAMS

4.10

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- Intermittent rivers and ephemeral streams (IRES) are ideal ecosystems in which to study how ecological and evolutionary processes interact across levels of organization, from genes to ecosystems
- Recurrent disturbance and succession processes are strong drivers of evolution in IRES and occur at varying spatial and temporal scales
- Species inhabiting IRES exhibit many adaptations that are subject to trade-offs related to flow intermittence
- Flow intermittence affects genetic and species diversity, speciation, and extinction, all of which affect ecosystem functions

4.10.1 INTRODUCTION

Despite occupying only 0.8% of the Earth's surface, freshwater ecosystems contain nearly 6% of all described species (Dudgeon et al., 2006). Understanding this diversity requires consideration of how ecology and evolution interact. Ecologists first recognized the importance of this interaction in the mid-20th century. Hutchinson's (1965) "Ecological theatre and evolutionary play" emphasized that evolutionary processes play out within a specific ecological context. Margalef (1959) also related ecology to evolution within the framework of ecological succession. He considered Hutchinson's ecological context to be highly dynamic because ecosystems are subject to disturbances that cause succession which then guide evolutionary processes. Margalef (1959) concluded that disturbances are key elements promoting the evolution of species.

Given that intermittent rivers and ephemeral streams (IRES) are characterized by flow cessation and, often, drying events that cause disturbance (here defined as events that reset biological communities; Williams and Hynes, 1977; Boulton, 2003), drying can be considered a selection pressure that gives rise to adaptations that allow organisms to cope with drying. IRES consist of habitat mosaics that disconnect and reconnect in both time and space (Datry et al., 2016a, Chapter 2.3). This dynamic landscape exerts strong controls on the population and community structure of aquatic organisms and on ecosystem functions (EFs). Therefore, IRES are ideal ecosystems in which to study how ecological and evolutionary processes interact.

Evolutionary ecology is the study of how ecological and evolutionary processes interact, with consequences across organizational levels from the individual to the biosphere (Fig. 4.10.1). Following,

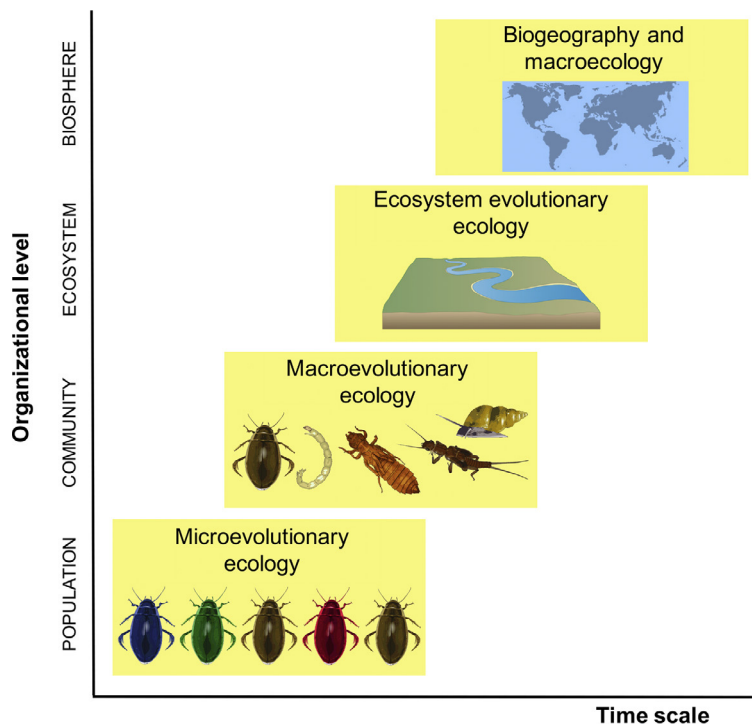


FIG. 4.10.1

Interactions between evolution and ecology have relevance for IRES across spatial and temporal scales and organizational levels. Microevolutionary ecology focuses on individual variability, here depicted with different colors of diving beetles (Dytiscidae). Macroevolutionary ecology focuses on species variability, here depicted with different invertebrate taxa (left to right: the diving beetle Dytiscidae, the midge Chironomidae, the dragonfly Gomphidae, the stonefly Nemouridae, and the snail Hydrobiidae; aquatic invertebrate drawings by Pau Fortuño). Ecosystem evolutionary ecology focuses on processes acting at the ecosystem scale, here represented with a meandering reach (symbol “ian-symbol-river-3d-meandering-with-sandy-banks” designed by Tracey Saxby, courtesy of the Integration and Application Network, University of Maryland Center for Environmental Science (ian.umces.edu/imagelibrary/)). Biogeography and macroecology focus on patterns that range from large regions to the biosphere.

we address genetic, evolutionary, and biogeographical/macroecological processes in IRES by analyzing the link between ecology and evolution at different levels. At the population level, we consider how IRES may influence selection, adaptation, and potential trade-offs, and how IRES spatiotemporal dynamics influence genetic diversity. At the community level, we consider how IRES promote niche differentiation and diversification, how trait variability can modify EFs, and how ecosystem characteristics influence adaptive evolution. At the biogeographical/macroecological level, we discuss the relative importance of local and regional factors and how speciation, extinction, and immigration can determine global diversity patterns. Our aim is to stimulate future research in IRES by adopting an evolutionary ecology approach. An overall goal is to emphasize that IRES have much to contribute to the field of evolutionary ecology.

4.10.2 MICROEVOLUTIONARY ECOLOGY PROCESSES IN IRES

Microevolutionary ecology focuses on individual- and population-level changes that result in adaptations to the environment. A centerpiece of microevolutionary ecology is the phenotype, or an observable trait, that directly or indirectly affects individual fitness. Traits are subject to intrinsic trade-offs because natural selection cannot optimize all traits simultaneously because resources are limited. Natural selection, mutation, genetic drift, gene flow, and sexual selection are the mechanisms of evolution, and along with fitness, adaptation, and phenotypic plasticity, are key concepts that apply to IRES (Box 4.10.1 and Section 4.10.2). Fig. 4.10.2 highlights open questions and hypotheses concerning the importance of each evolutionary process in each hydrological phase.

CHARACTERISTIC TRAITS TO COPE WITH LIFE IN IRES

Individuals respond to changing environmental conditions by dispersing, withstanding the event, or expressing phenotypic plasticity, all of which play a role in biotic responses to dry-phase disturbances. Drying can be a predictable (e.g., seasonal) or nonpredictable disturbance in the annual cycle of IRES. Although any repeated disturbance can lead to adaptive evolution, theory suggests that trait evolution will be particularly rapid when disturbances are frequent and/or predictable (Lytle et al., 2008).

Organisms can adapt to drying via behavioral, life history, and morphological changes (Lytle and Poff, 2004; Lytle, 2008) (Fig. 4.10.3; Chapters 4.7 and 4.8). Behavioral adaptations utilize environmental cues correlated with drying (e.g., increasing pool temperatures, decreasing water levels; Velasco and Millan, 1998) as a signal to escape by dispersing within a river or leaving the river entirely. Dispersal within the river network is important for highly mobile taxa such as fishes (Kerezy et al., 2013; Chapter 4.5) and some larger-bodied invertebrates (Boersma and Lytle, 2014; Chapter 4.3). Dispersal out of the river is especially important for mobile, air-breathing taxa such as adults of aquatic Coleoptera and Hemiptera. Life history adaptations are expected in taxa that cannot respond immediately to drying with behavioral avoidance or dispersal, but that may be able to anticipate drying from seasonal cues. For these organisms, phenology is closely tied to recurring seasonal drying (e.g., synchronous emergence; Cover et al., 2015).

Another example is diapause, common in arthropods but also occurring in some fishes such as Cyprinodontiformes (Chapter 4.5). Like dispersal that spreads mortality risk through space, diapause can be thought of a risk-spreading strategy that operates through time. Resting stages can occur at all life stages and persist for varying lengths of time from single dry seasons (e.g., the stonefly *Mesocapnia arizonensis*

BOX 4.10.1 DEFINITIONS OF KEY EVOLUTIONARY FORCES AND TERMS, EMPHASIZING THEIR RELEVANCE IN STUDIES OF IRES

One theme of this chapter is the implications of evolutionary processes in IRES. Central among these processes is *natural selection*, which winnows within-population trait variation while simultaneously adapting populations to their local environments. In IRES, natural selection is primarily imposed by the river's flow regime, particularly the predictability, timing, and duration of high, low, and zero flow events. *Genetic drift* is a population-level process that results from random fluctuations in gene frequencies, particularly in small populations such as isolated headwater populations or populations in disconnected dryland habitats. Small isolated populations are also vulnerable to inbreeding depression and extinction. *Dispersal (gene flow)* can “rescue” such populations via the influx of genetic variation (genetic rescue) or through the numerical addition of migrants (demographic rescue) (reviewed in [Carlson et al., 2014](#)). In IRES, dispersal among populations is an important mechanism by which many organisms escape drought conditions by moving to perennial refuges. Once flow resumes, dispersal from refuges to recently rewetted areas facilitates population recovery. *Mutation* is the generation of novel genetic variation *in situ*, which creates fodder for selection. Beyond these key evolutionary processes, we also define various terms from evolutionary ecology that appear throughout this chapter.

	Definition	Relevance in IRES
Absolute Fitness (population level)	A population-level parameter reflecting the average number of surviving offspring produced per capita per time period (i.e., the λ growth parameter).	Population models can be used to estimate the growth parameter over successive cycles of drying and rewetting phases in IRES.
Adaptation	A trait that confers increased fitness in an environment. An adaptation is an evolutionary response to selection.	Adaptations in IRES include resistance (e.g., diapause) and resilience traits (e.g., dispersal) that allow organisms to cope with specific drying and/or flowing conditions.
Dispersal	Involves the movement of individuals from one area (region, population) to another. In population genetics, migration indicates gene flow that results from movement between populations.	In IRES, dispersal to perennial refuges is a critical mechanism by which many organisms avoid desiccation during dry phases. Once flow resumes, dispersal then allows recolonization of rewetted habitats.
Fitness (Individual)	An individual's reproductive success, often estimated via fitness components such as survival, fecundity, or mating success. Individuals with higher fitness relative to others will be favored by natural selection.	Fitness estimates in IRES can vary greatly depending on the hydrological phase (dry vs. wet) and must be considered in study designs.
Genetic Drift	A population-level phenomenon that results from the random fluctuations in allele frequency owing to small population size.	Genetic drift is strongest in small isolated populations. Thus, drift may be pronounced during drying in IRES when populations are highly fragmented.
Isolation-by-Distance (IBD)	A pattern of increasing genetic distance among populations with increasing geographic distance	An example of IBD in IRES is provided in Section 4.10.2 (Fig. 4.10.6) .
Mutation	A physical/chemical change in DNA that occurs at the individual level. Mutations generate variation on which natural selection can act.	Mutation in IRES populations is one means of increasing genetic diversity, thereby creating variation on which selection can act. Mutation rates increase with stress, which may be caused by abrupt changes in environmental conditions (e.g., sudden loss of water or flash flooding in IRES).

BOX 4.10.1 DEFINITIONS OF KEY EVOLUTIONARY FORCES AND TERMS, EMPHASIZING THEIR RELEVANCE IN STUDIES OF IRES (Cont'd)

Natural Selection	The differential survival or reproductive success of individuals with different phenotypes. Natural selection acts on individuals, but it is measured as a change in the phenotype distribution within a generation. Selection can occur at different stages in an individual's life cycle.	In IRES, natural selection is imposed by the flow regime, particularly the timing and duration of high, low, and zero flow events.
Neutral Molecular Markers	Genetic markers for which mutational changes do not have any direct effect on fitness (i.e., selectively neutral).	Neutral molecular markers are often used—including in studies of IRES—to study dispersal and gene flow among populations.
Phenotypic Plasticity	The ability to express alternate phenotypes under different environments.	Plasticity may be common in IRES, where individuals adjust their growth and/or development in response to changing environmental conditions.
Sexual Selection	A special case of natural selection in which the traits under selection influence an individual's ability to acquire mates and reproduce.	Similar to natural selection, in IRES sexual selection is imposed by the flow regime, particularly the timing and duration of flow (and zero flow) events and how these factors influence mate acquisition and reproduction.

Gray, 1981) to multiple dry years (e.g., some microcrustaceans and rotifers, Jenkins and Boulton, 2007; Chapters 4.3 and 4.8). Morphological adaptations to drying include diapausing eggs, osmoregulatory structures, thickened cuticles, and other anatomical structures that confer desiccation resistance (Chapter 4.8). Fish species that estivate during the dry season can undergo metabolic suppression (African lungfish *Protopterus aethiopicus*, Delaney et al., 1974), possess an elongate body form for burrowing or maneuvering through small spaces (loach *Cobitis shikokuensis*, Kawanishi et al., 2013), or secrete mucus (West Australian salamanderfish *Lepidogalaxias salamandroides*, Allen and Berra, 1989; Chapter 4.5). In some cases, suites of traits may be favored in IRES and might produce a variety of adaptive landscapes (i.e., a variety of phenotypes or genotypes and their corresponding fitness options).

While many resistance and resilience traits are presumed to be adaptations to disturbance in IRES (Chapter 4.8), empirical studies of selection and adaptation in IRES remain rare. Selection (natural or sexual) is often studied in nature by examining trait shifts before and after some episode of selection. Understanding the causes of selection requires understanding the environmental circumstances that give rise to variation in fitness among individuals with different traits (Wade and Kalisz, 1990). An example from IRES is the caddisfly *Mesophylax aspersus* which avoids the dry season by estivating in nearby caves while eggs mature in the adult females. Salavert et al. (2011) found that natural selection favored large-bodied females during estivation, presumably because they had the energy reserves to succor their eggs and survive the estivation period.

Comparative studies provide important information and hypotheses, but conclusive studies of adaptation traditionally rely on common environment or reciprocal transplant experiments (Reznick and Travis, 1996). Lytle et al. (2008) quantified the flood-escape behavior of multiple populations of the

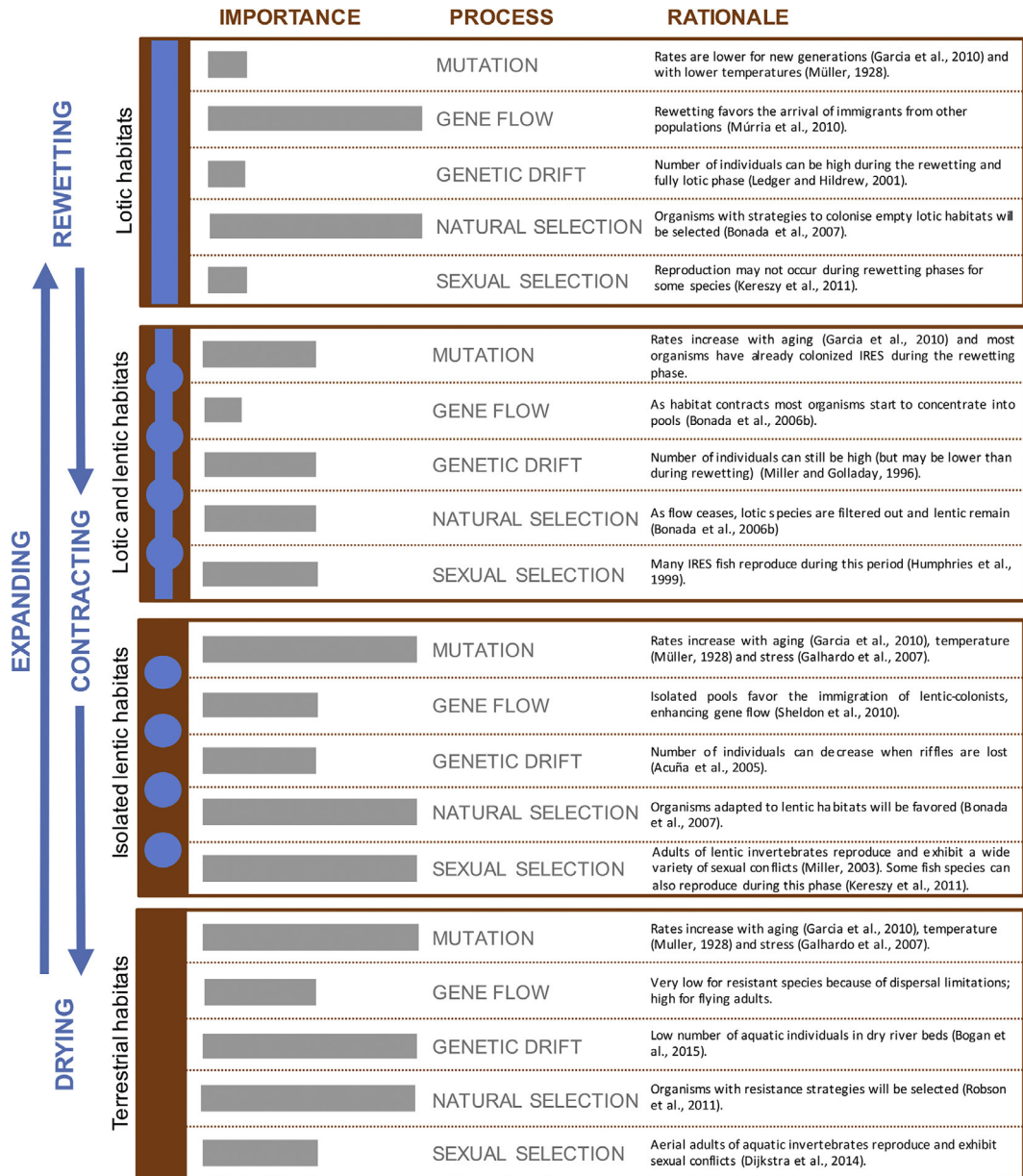
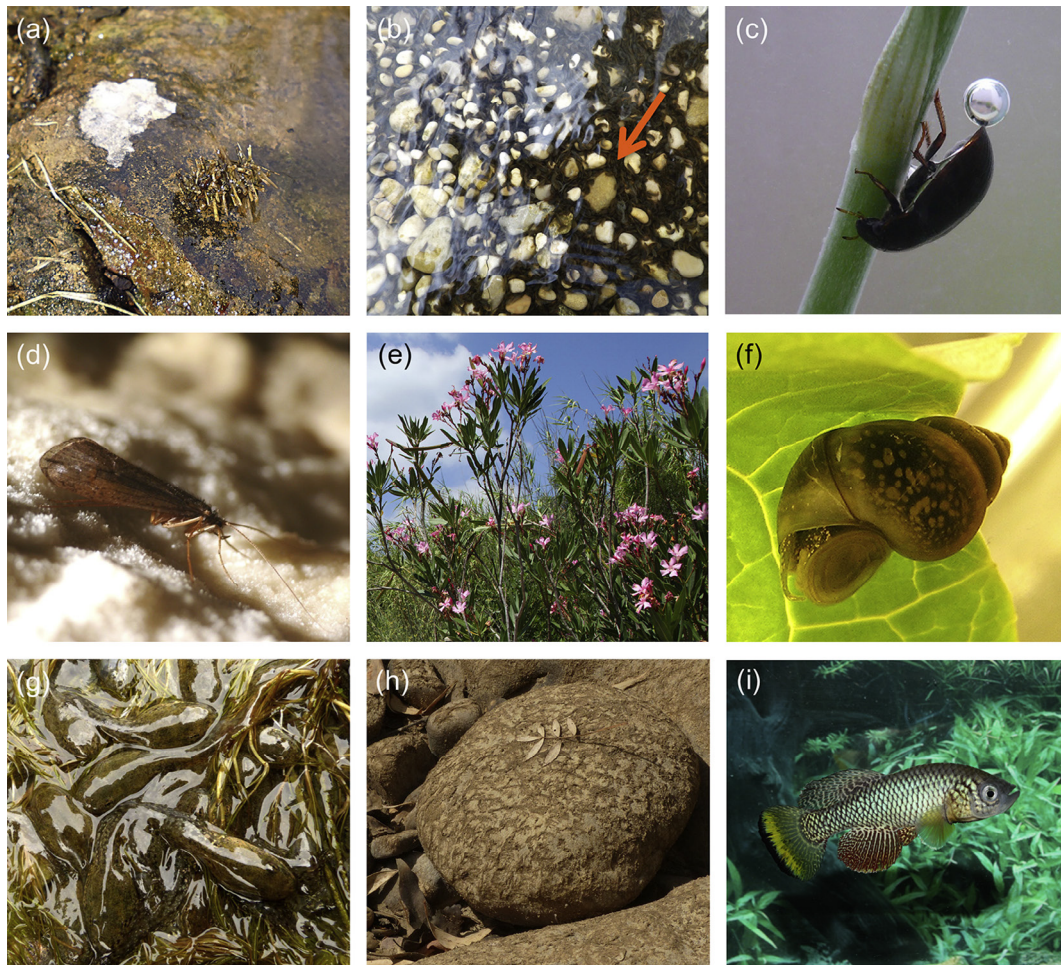


FIG. 4.10.2

Hypotheses about the relative importance (represented by the width of the gray bars) of the different evolutionary processes in each hydrological phase of an IRES, including our underlying rationale. Hydrological phases are presented from the rewetting phase where lotic habitats dominate to the contracting phase to the drying phase where terrestrial habitats dominate. Our rationale is focused on aquatic organisms; however, the importance of processes during the rewetting phase would be similar to those for terrestrial organisms during the drying phase. Blue rectangles indicate lotic habitats, blue circles indicate lentic habitats, and brown rectangles indicate terrestrial habitats. Arrows indicate contracting and expanding paths within wet and dry phases in IRES.

**FIG. 4.10.3**

Examples of IRES species with different traits to cope with drying: (a) the caddisfly *Limnephilus guadarramicus* has a stick case that increases resistance to drying; (b) some individuals, such as Gammaridae amphipod crustaceans (the brown patches indicated by an arrow), can move upstream or downstream through the aquatic habitat to find perennial refuges; (c) several species of aquatic beetles, such as *Hydrobius convexus*, can inhabit isolated pools during flow contraction by breathing aerial oxygen; (d) the caddisfly *Mesophylax aspersus* has diapausing adults that estivate in caves during dry phases; (e) the oleander *Nerium oleander* has leaves with stomata concentrated in crypts with trichomas that reduce transpiration; (f) many mollusks, such as *Bithynia tentaculata*, have an operculum that allow them to resist desiccation; (g) some amphibian species have tadpoles with life cycles synchronized with drying whereas others, such as those in this photo, are not; (h) many microbes and algae in biofilms can desiccate to withstand drying, reactivating upon rewetting; (i) the killifish *Nothobranchius furzeri* estivates during the dry season.

Photos: Courtesy A. Růfusová (a), N. Bonada (b,g and h), J. Arribas (c), T. Pérez-Fernández (d), T. Herrera (e), T. Machacek (f) and A. Furness (i).

flightless water bug *Abedus herberti* across a gradient of disturbance predictability, defined as the ability of a cue (rainfall) to predict a disturbance (flash flood). Common environment experiments, where populations from a range of environments were subject to the same conditions, revealed that individuals from predictable environments responded to rainfall by moving out of the water quickly, whereas individuals from less predictable environments did not. The study provides compelling evidence for adaptive divergence among populations in flood-escape behavior that correlates with the environmental predictability at their home sites. It also indicates that drying is not the only evolutionary driver in IRES and, due to cotolerance, many traits selected by other disturbances such as floods can favor the presence of some taxa over others (Vander Vorste et al., 2016).

Aquatic organisms also can alter life history strategies or behaviors to match a range of drying conditions through plasticity (Lytle, 2008). Insects in IRES benefit from plastic strategies that allow different trajectories of growth and development, contingent on whether or not a dry phase occurs (Johansson et al., 2001; Wissinger et al., 2003; De Block and Stoks, 2004, 2005). These strategies require sufficient proximate cues to signal whether or not a dry phase will occur. For populations studied during drying events, phenotypic plasticity does appear to be common; individuals adjust their rates of growth, development, or both to compensate for different environments. For instance, experiments with alpine caddisflies (Shama and Robinson, 2006) and damselflies (De Block and Stoks, 2004) found that plasticity, rather than local adaptation, accounted for most observed differences between populations from perennial rivers and IRES. Thus, it is possible that IRES may select for traits that facilitate phenotypic plasticity across environments that dry, rather than favoring traits focused solely on surviving dry conditions (Vander Vorste et al., 2016). The prevalence of phenotypic plasticity across taxa will depend largely on whether there exists a trade-off between plasticity itself versus specialization in environments prone to drying. Phenotypic plasticity has been associated with epigenetic mechanisms (i.e., a change in gene expression without a change in genotype). How environmental cues influence heritable epigenetic mechanisms is still an open question in many fields of biology (Duncan et al., 2014), and the environmental cues provided by drying offer a good opportunity to further investigate epigenetic mechanisms in IRES species.

POTENTIAL TRADE-OFFS IN IRES: TO STAY OR TO LEAVE?

Dispersing or remaining can be viewed as alternative strategies for spreading risk through space and time (Den Boer, 1968), but most studies focus on one, not both (Bohonak and Jenkins, 2003), limiting our ability to examine trade-offs between them. A trade-off should lead to negative covariation between the two strategies (e.g., organisms with complex life histories may disperse through space less than organisms with more simple life histories, Fig. 4.10.4), as predicted from theory. Although much of the existing empirical literature focuses on plants with dormancy capability (Venable and Lawlor, 1980), aquatic invertebrates with desiccation-resistant life stages hold considerable potential as an empirical study system for exploring such trade-offs (see also Bohonak and Jenkins, 2003). How the two strategies interact to influence the probability of persistence is an open question (Buoro and Carlson, 2014) with much relevance for the study of IRES.

GENETIC STRUCTURE IN IRES

Whether a species is connected or isolated depends on its dispersal ability and the spatial characteristics of flow intermittence and hydrological connectivity (Zickovich and Bohonak, 2007; Sheldon et al., 2010; Chester et al., 2015; Box 4.10.2 and Fig. 4.10.5). Phillipsen et al. (2015) studied the

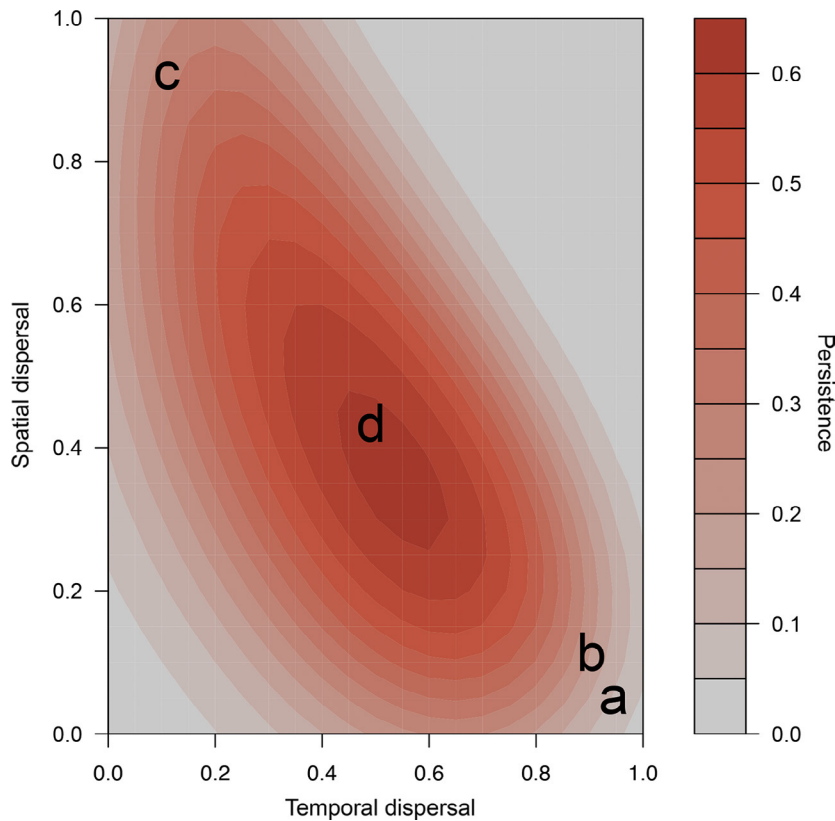


FIG. 4.10.4

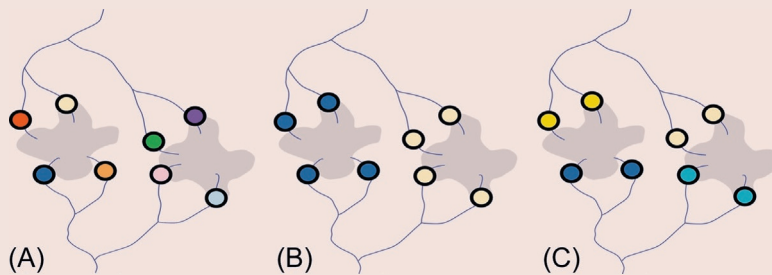
Hypothesized relationship between persistence probabilities at a site as a function of spatial and temporal dispersal strategies. Persistence probabilities were generated from simulations using a positive effect of both dispersal strategies on persistence, but a negative effect of interactions between the two (modified from [Buoro and Carlson, 2014](#)). Overlain on the figure are hypotheses about where certain species of IRES specialists fall along the curve: (a) the stonefly *Mesocapnia arizonensis* (Michael T. Bogan, personal communication) has a complex life history including dormancy, suggesting high potential for temporal dispersal, but males are effectively flightless, suggesting little opportunity for spatial dispersal at the species level; (b) the dobsonfly *Neohermes filicornis* (Matthew R. Cover, personal communication) also has a complex life history, including dormancy, but little is known about the spatial dispersal capabilities of this species; (c) the beetle *Boreonectes aequinoctialis* has high spatial dispersal capabilities (inferred from [Phillipsen et al., 2015](#)), (d) the caddisfly *Mesophylax aspersus* has adults that disperse several kilometers during the dry season and estivate in caves until flow resumes ([Salavert et al., 2008](#)), suggesting intermediate temporal and spatial dispersal in this species. Whether the observed differences lead to similar persistence probabilities remains unknown.

BOX 4.10.2 GENERAL MODELS OF GENE FLOW IN STREAMS

Death Valley Model (DVM) (Fig. 4.10.5A). Local populations exhibit little or no gene flow and, in some cases, evolve into species flocks of incipiently diversifying species. The model is named for its original application to describe genetic patterns among highly isolated spring populations in arid regions such as the Mojave Desert of the southwestern United States (Meffe and Vrijenhoek, 1988). The DVM might be highly applicable for low-dispersal species in IRES stream networks with high flow intermittence, where aquatic connectivity is frequently severed.

Headwater Model (HWM) (Fig. 4.10.5B). Headwater specialist taxa show more genetic affinity across short terrestrial distances than within the stream network itself. The HWM was originally described for species occupying the upper limits of predominantly intermittent stream networks and that have some ability to disperse terrestrially (Finn et al., 2007). The HWM is likely to fit many headwater taxa in IRES, especially aquatic insects that disperse terrestrially, because drought often disrupts aquatic connectivity of intervening reaches (Chester et al., 2015; Cañedo-Argüelles, et al., 2015; Razeng et al., 2016).

Stream Hierarchy Model (SHM) (Fig. 4.10.5C). Genetic connectivity is nested within the hierarchical structure of river stream networks (Meffe and Vrijenhoek, 1988). This is the pattern to be expected in species that occur throughout stream networks and that rarely or never disperse terrestrially, such as many fishes and some flightless invertebrates. In IRES, the SHM would be expected in systems with low flow intermittence, where surface flows connect local populations.

**FIG. 4.10.5**

General models of gene flow in streams (adapted from Hughes et al., 2009; Crook et al., 2015): the Death Valley Model (A), the Headwater Model (B), and the Stream Hierarchy Model (C).

Modified from Crook, D.A., Lowe, W.H., Allendorf, F.W., Erös, T., Finn, D.S., Gillanders, B.M., et al., 2015. Human effects on ecological connectivity in aquatic ecosystems: integrating scientific approaches to support management and mitigation, *Sci. Total Environ.* 534, 52–64, with permission from Elsevier.

population genetics of three species of aquatic insect inhabiting an IRES network in arid southeastern Arizona. The first, the giant water bug *A. herberti*, is restricted to perennial stream reaches and exhibited extreme between-population differentiation that was uncorrelated with geographic distance (Fig. 4.10.6). Significant divergence occurred between populations in the same stream, separated by less than 1 km of intermittent habitat, a pattern perhaps common in IRES species that require aquatic habitat (Fig. 4.10.2). The second, the stonefly *M. arizonensis*, which is only found in IRES reaches that sometimes lack surface water, exhibited an isolation-by-distance pattern, likely due to the ability of individuals to fly within and among streams. The water beetle *Boreonectes aequinoctialis* showed a panmictic population genetic pattern due to its ability to fly long distances and utilize nearly any surface water as habitat (Fig. 4.10.6). These different dispersal abilities of aquatic organisms and the

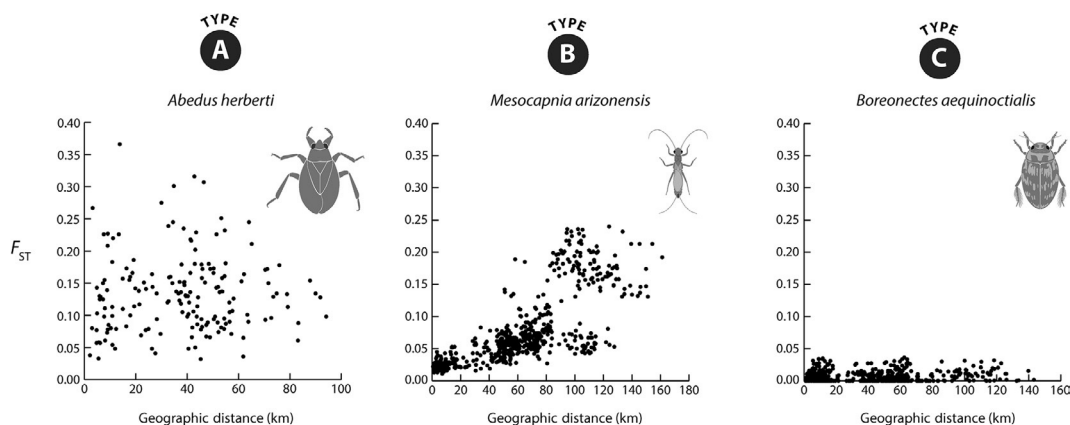


FIG. 4.10.6

The spatial structure of IRES produces distinctive population patterns, depending on the dispersal ability of the organism. F_{ST} is a measure of pairwise genetic difference between populations (i.e., higher the F_{ST} , the greater the distinction between populations). Type A shows a flightless perennial-water specialist, the giant water bug *Abedus herberti*. Populations are strongly differentiated due to genetic drift. Type B shows an intermittent-habitat specialist, the stonefly *Mesocapnia arizonensis*, that disperses short distances aerially. Populations exhibit an isolation-by-distance pattern where genetic distance increases with geographic distance. Type C shows a strong-flying habitat generalist found in IRES, the beetle *Boreonectes aequinoctialis*. Genetic structure is panmictic (i.e., random mating), with no apparent spatial pattern.

From Phillipsen, I.C., Kirk, E.H., Bogan, M.T., Mims, M.C., Olden, J.D., Lytle, D.A., 2015. Dispersal ability and habitat requirements determine landscape-level genetic patterns in desert aquatic insects. *Mol. Ecol.* 24, 54–69.

network configuration of flow intermittence influence the potential for metapopulation dynamics in IRES (Múrria et al., 2010; Datry et al., 2016a; Chapter 4.9).

4.10.3 MACROEVOLUTIONARY ECOLOGY PROCESSES IN IRES

Macroevolution focuses on the species and lineage levels, and aims to understand community ecology from an evolutionary perspective (Fig. 4.10.1). Many studies on IRES report that some species are IRES specialists whereas others are generalists (Leigh et al., 2016a; Chapter 4.3). However, most community studies of IRES lack an evolutionary perspective and, at best, only describe the biological traits of the species within a community (Bonada et al., 2007; Datry et al., 2014). Linking community ecology to evolutionary biology should provide a more theoretical and fundamental basis for understanding species diversification and assemblages. In this vein, Vellend (2010) suggested that similar rules apply to both population genetics and community ecology. For example, natural selection favors individuals with the highest fitness, while environmental filtering favors species with traits that better match the environment. Genetic drift in population genetics can be considered analogous to ecological drift in community ecology. Much like mutation is the ultimate source of genetic variation, speciation

is the ultimate source of species diversity. Finally, the ecological equivalent to gene flow in population ecology is dispersal in community ecology. All of these concepts are also fundamental to understanding communities within IRES.

Empirical and theoretical evidence suggests that local communities in streams are structured predominantly by a dynamic balance between the environmental characteristics of the local habitat (species sorting) and regional-scale dispersal and postdisturbance recolonization (mass effects) (Thompson and Townsend, 2006; Brown et al., 2011). Deterministic biotic interactions are thought to be less important drivers of community structure. In IRES, the repeated pattern of drying and rewetting suggests that recolonization by aquatic-obligate species is an essential process in structuring local communities, and that mass effects might be even more important in IRES than species sorting during particular hydrological phases (Datry et al., 2016b).

Aquatic communities in IRES, particularly those occupying the “harsh” end of the intermittence gradient (i.e., those spending a greater proportion of time without than with surface flow), tend to comprise predictable subsets of the regional metacommunity that have high-dispersal ability (Datry et al., 2014; Chapter 4.9). This pattern suggests a sink-type nature of IRES, in which the local communities rely on connectivity with communities in more stable streams that act as recolonization sources (Fig. 4.10.7). Therefore, well-connected metacommunities can be said to confer resilience. However,

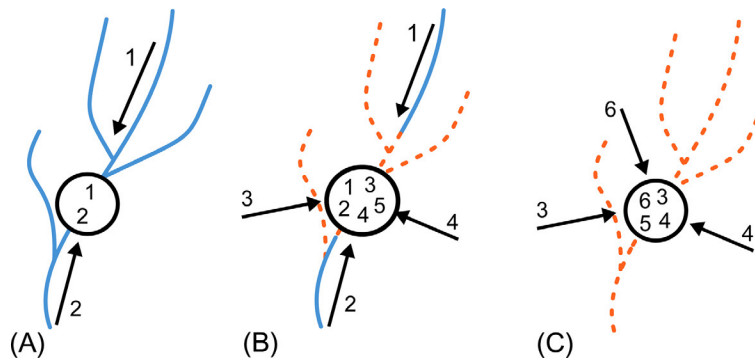


FIG. 4.10.7

Trait variability of colonizing individuals/species in a perennial stream (A) and two IRES networks with different degrees of flow permanence (B and C). Circles represent particular sites and the *solid blue* and *dashed orange* lines represent perennial and intermittent river sections, respectively. Arrows represent the origin of colonizing individuals/species, and numbers represent the genetic configuration of colonizing organisms. In the perennial network (A), individuals/species of a particular site have the trait configuration of the individuals/species arriving from upstream and downstream permanent sites. Arrival of individuals/species from nearby sites is likely to be limited (i.e., habitat is already occupied). In the IRES network with intermediate levels of flow permanence (B), there is an increase of trait variability in populations/communities following flow resumption because individual/species arrive from upstream, downstream, and nearby locations. In the IRES network with low levels of flow permanence (C), trait variability decreases during flow contraction but these systems can still have relatively high levels of trait variability due to immigration of individuals/species with different trait values arriving randomly from nearby locations. In both IRES examples (B and C), the number 5 refers to characteristic IRES individuals/species with resistance forms that are present year-round.

growing evidence suggests that a rarer component of IRES communities is resistant to the inherently harsh wet-dry alternations (Chapter 4.8) and remains in place (Fig. 4.10.7). This component includes the estivating *M. arizonensis* described earlier (Phillipsen et al., 2015). The “low-disperser” component of the metacommunity that includes *M. arizonensis* shows a pattern of distance decay of regional community similarity (Cañedo-Argüelles et al., 2015). This spatial pattern is analogous to the pattern of isolation-by-distance among populations of *M. arizonensis* shown in Fig. 4.10.6. At the community level, this pattern reflects the neutral effects of dispersal limitation across the landscape (e.g., Thompson and Townsend, 2006) and suggests that evolution *in situ* could be more likely in low-dispersal species of the full assemblage than in high-dispersal species.

Resistance is important for macroevolutionary processes such as speciation because diversification is expected to proceed more rapidly in less dispersive lineages that are more likely to remain in a local habitat. Research comparing lotic and lentic environments has found lentic species to have larger ranges (Hof et al., 2006; Abellán and Ribera, 2011), steeper species accumulation curves (Ribera et al., 2003), and lower population differentiation (Hjalmarsson et al., 2015; Marten et al., 2006) than lotic species. The hypothesis is that the comparatively short geological life span of standing water bodies requires lentic lineages to be more dispersive than lotic lineages (Ribera, 2008). Perhaps IRES and lentic systems act similarly, albeit at shorter time scales for IRES. The only explicit test of habitat stability on diversification in IRES found no significant link to habitat (Dijkstra et al., 2014), but this hypothesis warrants further investigation.

In the absence of significant spatial dispersal, species that are resistant to drying have the potential to diversify in allopatry (i.e., while separated geographically). Allopatric diversification can be driven by different selective pressures in different localities (adaptive divergence, Lytle et al., 2008; Watanabe et al., 2014), by neutral genetic drift in isolation or by a combination of the two. Furthermore, neutral drift leading to diversification can result from spatial or temporal isolation. Genetic divergence at putatively neutral markers (e.g., microsatellites) can also occur when dispersal between localities is low enough to permit populations to diverge, even without strong selective forces (Finn et al., 2006, 2007). A developing idea in stream ecology that is likely to be highly relevant to IRES is that even species with moderate dispersal ability that have populations with minimal spatial isolation can evolve in divergent directions if environmental conditions vary among localities (Watanabe et al., 2014). One important example is when conditions lead to significant differences in the timing of key life history events. For example, mayfly populations occupying nearby alpine streams fed by either glacier meltwater or groundwater reach maturity at significantly different times of the season due to the extremely different thermal and hydrological regimes of these two stream types (Finn et al., 2014). Boumans (2014) also found strong evidence for divergence/speciation in stoneflies occupying sites with different hydrological regimes that influenced the timing of adult emergence. Such evolutionary diversification via separation in time rather than space has been termed “temporal allopatry” (Boumans, 2014) or temporal “reproductive isolation” (Finn et al., 2014). Although not widely studied in aquatic insects (see Dijkstra et al., 2014), temporal variation in the durations of drying in IRES may explain closely related species that are distributed longitudinally along river systems (Mey, 2003) or the co-occurrence of distinct lineages within recognized species in sympatry (Leys et al., 2014). Because IRES produce brief windows of opportunity for life history events, and because the timing is likely to vary, IRES provide multiple opportunities to investigate the degree to which such temporal allopatry might be driving evolutionary diversification and, eventually, speciation.

4.10.4 EVOLUTIONARY PROCESSES AT THE ECOSYSTEM LEVEL IN IRES

Ecosystem ecologists study processes that occur at the ecosystem level, such as fluxes of matter (including organisms) and energy. They consider individuals and species as part of the ecosystem, but only recently have started to analyze how evolutionary differences among them (i.e., genetic and species diversity, respectively) determine EFs (Matthews et al., 2011). Each individual and species plays a particular functional role in an ecosystem, and even closely related individuals or species can have functional differences. For instance, several closely related species of the caddisfly genus *Drusus* have different feeding habits (grazers, shredders, or predators) acquired through evolution, thus contributing differently to the organic matter processing of river ecosystems (Pauls et al., 2008). Understanding how levels of genetic and species diversity affect EFs is where evolution and ecosystem ecology meet. Obviously, EFs not only depend on the genetic and species diversity in the ecosystem but also on present environmental factors such as temperature and light. Disentangling evolutionary and environmental effects involved in EFs helps to tease apart purely evolutionary effects from environmentally related effects.

BIOLOGICAL TRAITS IN IRES AND THEIR RELATION TO ECOSYSTEM FUNCTIONS

The functional role of an individual or species in an ecosystem is expressed through their biological traits (Violle et al., 2007). To understand evolutionary effects at the ecosystem level in IRES, a comprehensive knowledge of traits related to EFs is needed (Table 4.10.1). For aquatic organisms inhabiting IRES, traits such as feeding habits, body size, life cycle characteristics, and behavior have been related to EFs (Bonada et al., 2006a). Feeding habits influence organic matter processing, whereas body size and life cycle duration are related to ecosystem productivity. Especially relevant in IRES are life cycle characteristics, dispersal abilities, or strategies to withstand drying, and these are indirectly related to ecosystem resilience or resistance (i.e., the capacity of an ecosystem to respond to a disturbance, Datry et al., 2014; Chapter 4.8).

Traits related to EFs can be considered neutral or nonneutral in terms of their influence on ecosystem performance. Neutral traits would not change EFs under specific environmental conditions, whereas nonneutral traits would be favored by natural selection. Following flow resumption in IRES, individuals and species with traits that favor colonization (e.g., higher dispersal abilities, short life cycles) would be positively selected, enhancing ecosystem resilience. Other traits would be neutral for resilience following flow resumption. For instance, being a shredder or not would not specifically affect an organism's resilience to drought.

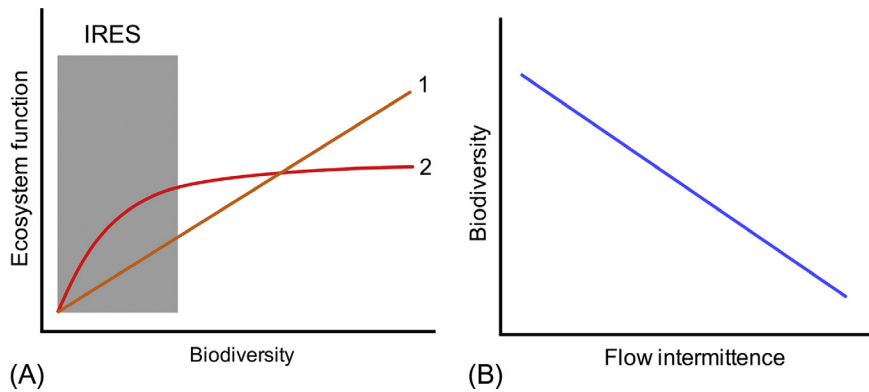
BIODIVERSITY-ECOSYSTEM FUNCTIONS RELATIONSHIPS IN IRES

Disturbance strongly influences genetic and species diversity (Banks et al., 2013). Given the constant presence of human disturbances, there is growing concern about the functional consequences of biodiversity loss. Scientists are now striving to understand the relationships between biodiversity and EFs. Overall, there is a positive relationship between biodiversity and ecosystem performance (see Srivastava and Vellend, 2005; Fig. 4.10.8A) and this relationship applies to species and genetic diversity (Whitham et al., 2006). The biodiversity-EF relationship is hypothesized to be highly dynamic in space and time in IRES. Flow intermittence decreases species (Datry et al., 2014; Fig. 4.10.8B) and

Biological trait	Related ecosystem function	Examples
Resources supply	Organic matter production	Algae, macrophytes, and riparian vegetation produce organic matter through photosynthesis
	Organic matter processing	Shredders process organic matter accumulated in river beds
	Carbon sequestration	Autotrophic organisms (algae, macrophytes, and riparian vegetation) capture and store carbon dioxide
	Carbon emission	Heterotrophic organisms (microbes, invertebrates, vertebrates) release carbon dioxide through their metabolism
	Nutrient cycling	Through their metabolism, fish excrete nitrogen and phosphorus
	Food web structure	Feeding habits of species determine food web structure
	Food web stability	Diversity of feeding habits of species determines food web stability
Body size	Biomass	Large fish can provide larger amounts of food resources for top predators
	Productivity	Large species have been traditionally associated with lower productivity values
Dispersal ability	Stability	Large consumers can increase ecosystem stability
	Resilience	Species with high dispersal ability will rapidly colonize new empty habitats
Life history	Connectivity	Species with high dispersal ability will increase population connectivity and increase the probability of persistence
	Production	Species with short life cycles and several generations per year increase production
	Resilience	Species with short life cycles and several generations per year increase ecosystem resilience
Behavior characteristics	Resilience	Species with large numbers of offspring increase ecosystem resilience
	Engineering	Burrowing organisms promote vertical connectivity and nutrient cycling
	Pollination	Plant species with higher ability to mobilize gametes increase pollination
	Retention of solids	Species that build cases with mineral substrates contribute to retention of solids

genetic diversity (Meffe and Vrijenhoek, 1988; but see Zickovich and Bohonak, 2007). According to the biodiversity-EF relationship, low diversity in IRES will reduce EFs (Fig. 4.10.8A and B). However, the few studies exploring this relationship in IRES have found substantial functional redundancy (Schriever et al., 2015; Vander Vorste et al., 2016), indicating that even if species diversity decreases, EF can remain relatively stable.

The genetic diversity-EF relationship has been little studied in freshwaters. The few existing examples from terrestrial and marine ecosystems (e.g., Reusch et al., 2005; Kettenring et al., 2014) indicate that genetic diversity increases EFs. We propose that during nonflowing phases, only individuals with

**FIG. 4.10.8**

The relationship between biodiversity and ecosystem function (A) according to two general models (1: linear and 2: log-linear) (Srivastava and Vellend, 2005). IRES (represented here with a gray box) are expected to have lower ecosystem function than perennial rivers because of the negative relationship between biodiversity and flow intermittence (B) (Datry et al., 2014).

resistance to drying will sustain EFs (Roger et al., 2012). However, these individuals are expected to have lower genetic variability, with genotypes highly adapted to the local environment and with high levels of isolation because of their limited dispersal. Several fish species from isolated populations in desert IRES, for instance, show low genetic diversity within populations and high genetic divergence among populations (Meffe and Vrijenhoek, 1988).

IRES are ecosystems with highly dynamic metapopulations and metacommunities (Chapter 4.9) where local adaptation to flow intermittence (the resting individuals or species) and dispersal (the vagile individuals or species) can regulate EFs. Colonization sources in IRES can be very diverse (Datry et al., 2016b). Following flow resumption or during the nonflowing phase, individuals recolonize from upstream or downstream perennial reaches or from other reaches in nearby catchments (Datry et al., 2016b; Chapters 4.8 and 4.9). Alternatively, they may display a high degree of within-reach affinity. Many of the vagile individuals will randomly arrive from all these sources, but only those with favorable traits will succeed. These surviving individuals will potentially increase the individual trait variability and thus enhance the suite of EFs (Fig. 4.10.7). More research is needed to more fully understand how genetic diversity in these highly dynamic systems affects EFs (Chapter 4.9).

RECURRENT ECOLOGICAL SUCCESSION IN IRES AS A FRAMEWORK FOR FAST EVOLUTION

Flow resumption in IRES transforms terrestrial habitats to aquatic ones, flow cessation extinguishes lotic habitats, and drying transforms aquatic habitats to terrestrial ones. All these events open new spaces for succession to proceed and for evolution to act. Several sets of ecosystem properties change as ecological succession proceeds. Complexity, biomass, trophic levels, specialization, and coevolution increase, whereas the production:biomass ratio and irreversibility decrease (Margalef, 1959; Fig. 4.10.9). These changes are thought to be universal rules of succession and imply that conditions for natural

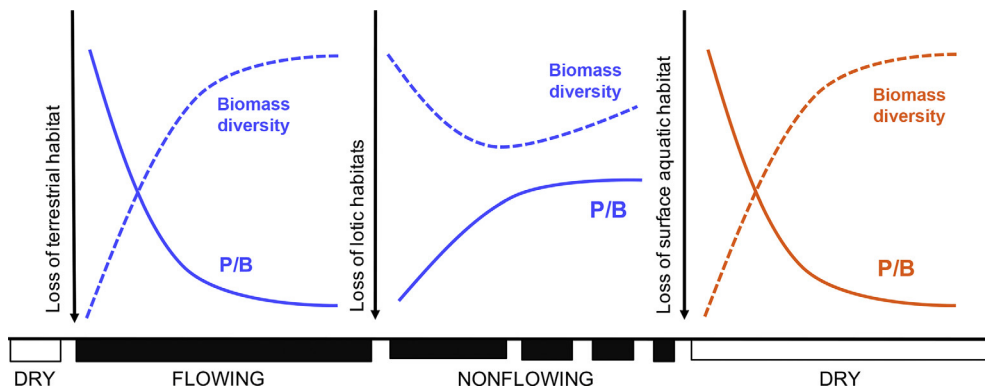


FIG. 4.10.9

Application of [Margalef's \(1959, 1986\)](#) ideas on the relationship between evolution and ecological succession in IRES. The x-axis symbolizes the changes in hydrological phases over time. Vertical arrows symbolize critical moments for biota and ecosystem functions. Blue lines correspond to aquatic organisms and orange lines to terrestrial organisms. Following flow resumption, the biomass and diversity of aquatic organisms increase whereas productivity (P/B: Production/Biomass) decreases. With the loss of lotic habitats, the biomass, diversity and productivity of aquatic organisms is also expected to decrease until communities collapse due to the loss of surface aquatic habitat. Finally, the colonization of dry riverbeds results in an increase of the biomass and diversity of terrestrial organisms and a decrease in their productivity.

selection to act should also change as succession proceeds ([Margalef, 1959](#)). Only species adapted to the next successional stage of these universal changes pass through the environmental filter and are favored by natural selection. Ecosystems are recurrently subject to disturbances and adaptive evolution should therefore be guided by successional rules ([Margalef, 1986](#)).

Recurrent ecological succession can lead to rapid evolution in IRES. Evidence supporting this idea comes from the numerous adaptations that exist in both aquatic and terrestrial species of IRES to cope with flow cessation and resumption conditions. The loss of riffles in IRES represents a tipping point for aquatic communities ([Chapters 2.2, 2.3, and 4.9](#)). Lotic species disappear, reducing biomass and diversity but lentic species that slowly increase in biomass, diversity, and productivity are favored ([Fig. 4.10.9](#)). These include species with few offspring, low-dispersal abilities, or long life cycles ([Margalef, 1959](#)). Many dragonflies, beetles, and true bugs inhabiting IRES when riffles are lost have some of these characteristics ([Bonada et al., 2007; Chapter 4.3](#)). As habitat contraction proceeds, pools eventually disappear and a newly available habitat—the terrestrial habitat—appears. Aquatic species disperse to more suitable habitats, become locally extinct, or become part of the “seed bank” ([Chapter 4.3](#)), whereas terrestrial species progressively colonize the dry riverbed ([Steward et al., 2012; Chapter 4.4](#)). In turn, terrestrial species present at these initial stages would be, again, opportunistic according to the succession rules, peaking in productivity and slowly increasing in biomass and diversity with time ([Fig. 4.10.9](#)).

Flow resumption can also promote rapid evolution in terrestrial and aquatic species. Numerous terrestrial invertebrates may be entrained by flow; some of these die, while others escape from inundation by displaying different behaviors ([Corti and Detry, 2012; Chapter 4.3](#)). In contrast, aquatic species with numerous offspring, high-dispersal abilities, short life cycles, and successful competition relative

to congeneric species are favored. The evolutionary rate of these aquatic species (i.e., changes across many generations) is expected to be fast and species that are not opportunistic would most likely not be present (Margalef, 1959). However, the presence of resting species in IRES that survive the dry period in the “seed bank” (Chapter 4.3) may also contribute to succession and affect the evolution of colonizing species by adding an extra constraint (i.e., biotic interactions). Traits that favor colonization and reduce biotic interactions with already existing species would be favored by natural selection. Species possessing these favored traits will increase productivity at the beginning of the flow resumption period, whereas biomass and diversity will slowly increase with time (Fig. 4.10.9).

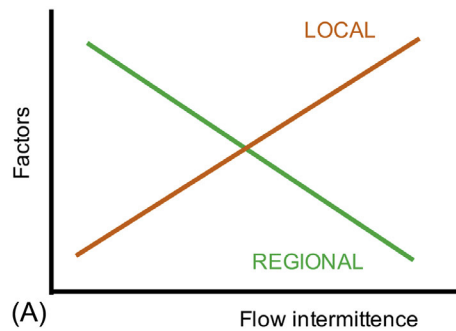
4.10.5 GLOBAL BIOGEOGRAPHICAL AND MACROECOLOGICAL PATTERNS OF AQUATIC ORGANISMS IN IRES

IRES have received little attention in biogeographical and macroecological studies of aquatic organisms (but see Leigh and Datry, 2016). This is surprising considering the global ubiquity of IRES (Chapter 1) and the increase in their ecological investigation in the last decade (Datry et al., 2011; Leigh et al., 2016a). A precise global inventory of IRES and their characteristics (i.e., duration, frequency, and magnitude of drying) is needed for a macroecological evaluation of the diversity they sustain. As more temporal and spatial water flow data become available and hydrological models more accurate (Chapter 2.2), this research gap should be filled.

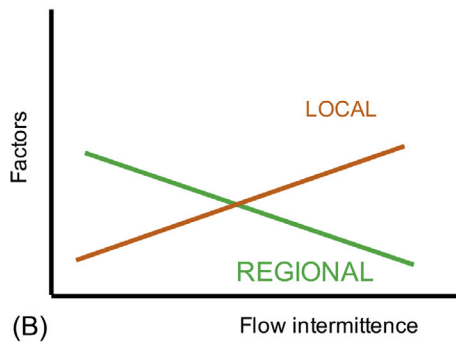
Global biogeographical and macroecological studies of freshwater diversity have included organisms adapted to both permanent and intermittent freshwater habitats. The characteristics of IRES certainly contribute to current diversity patterns across different regions. Flow-related environmental variables in global or continental freshwater biodiversity assessments give, at best, a broad perspective of the importance of the hydrological regime for freshwater diversity (e.g., Tisseuil et al., 2013), basically spreading a message of “less flow-less diversity.” This certainly applies to IRES aquatic diversity which is negatively related to flow intermittence (Datry et al., 2014).

LOCAL AND REGIONAL FACTORS DETERMINING DIVERSITY IN IRES

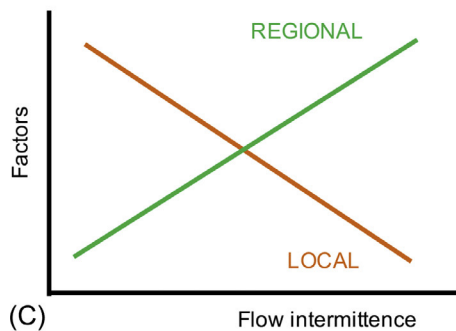
The study of biogeography and macroecology considers large-scale drivers of diversity as affected by factors of habitat-size (i.e., the species-area effect), resource availability (i.e., the species-energy effect), and historical contingencies (e.g., impacts of major past climate changes like the Pleistocene glaciations). Given the strong influence of flow intermittence on diversity and nestedness of aquatic communities, local factors such as the distribution of aquatic refuges across river landscapes or the dispersal abilities of IRES species (Chapters 4.8 and 4.9) may be important in IRES with high levels of intermittence (Leigh and Datry, 2016), whereas area, climate, and latitude may be more important in IRES with low levels of flow intermittence and with surface water most of the year. Razeng et al. (2016) found that the community composition of aquatic invertebrates from central Australia, a region where IRES dominate the landscape, was influenced primarily by topographic connectivity via overland dispersal pathways. However, predictions may differ between strictly and nonstrictly aquatic organisms because they have different dispersal capacities (Fig. 4.10.10). For strictly aquatic organisms (e.g., fish, some aquatic invertebrates) that have a single pathway to colonize IRES from perennial stretches or refuges, local factors driving diversity may be more relevant than for nonstrictly aquatic



Strictly aquatic



Nonstrictly aquatic



Terrestrial



FIG. 4.10.10

Hypothesized assumptions about the relative importance of local (in orange) and regional (in green) factors in explaining diversity in IRES of organisms that are: (A) strictly aquatic such as the fish *Squalius malacitanus* and the mollusk *Unio* sp., (B) not strictly aquatic such as the veliid bug *Velia caprai caprai* and the caddisfly *Limnephilus guadarramicus*, and (C) terrestrial such as the wolf spider *Pardosa saltans* and the rove beetle *Paederidus ruficollis*.

Photos: Courtesy T. Herrera - the fish, mollusk and veliid bug; J. Ortiz (Associació CEN) - the caddisfly; and R. Corti - the spider and rove beetle.

organisms (e.g., those invertebrates that can fly) at similar levels of intermittence (Cañedo-Argüelles et al., 2015; Detry et al., 2016b). For terrestrial organisms (i.e., those colonizing dry river beds), regional factors may become more important as flow intermittence increases in duration and riverbeds remain dry for longer periods of time (Fig. 4.10.10).

SPECIATION, EXTINCTION, AND IMMIGRATION IN IRES

IRES often comprise the headwaters of river networks (Benstead and Leigh, 2012) where local diversity is usually low (Finn et al., 2011). Compared to terrestrial landscapes, where individuals move through several dispersal routes, the movements of aquatic organisms within drainage basins (especially for strictly aquatic organisms) are restricted along the aquatic branches of dendritic river networks (Benda et al., 2004). In IRES, the naturally higher isolation of headwaters combined with their intermittence should have important consequences for the mechanisms driving speciation, extinction, and immigration (Fig. 4.10.11). As a consequence, and because isolation and fragmentation greatly influence genetic divergence of aquatic organisms (Burridge et al., 2006; see Section 4.10.2), different levels of intermittence are likely to affect these mechanisms and contribute to large-scale diversity gradients

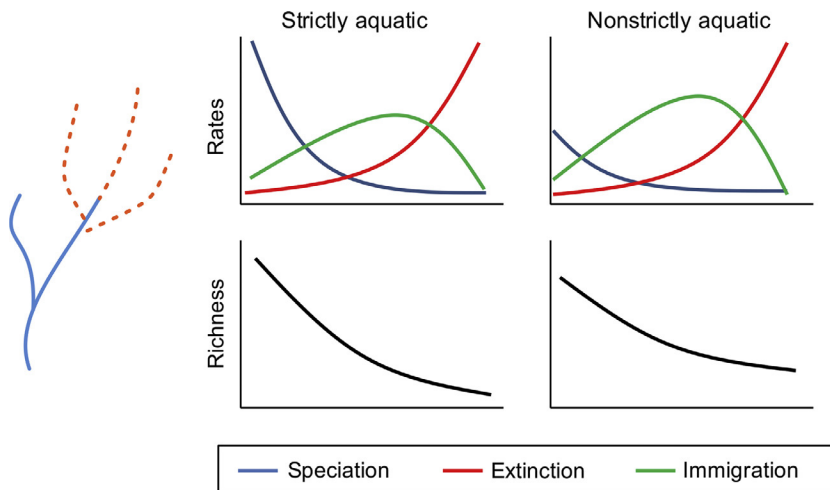


FIG. 4.10.11

Assumptions about evolutionary processes shaping diversity patterns in headwater streams along a gradient of flow intermittence (horizontal axes) for strictly and nonstrictly aquatic organisms, assuming perennial flow in downstream sections provides a constant source of organisms. The decreasing pattern of richness along this flow-intermittence gradient would be mainly driven by speciation at low levels of intermittence, then giving way to communities supported by immigration from downstream permanent reaches before finally being outweighed by extinction rates at high levels of intermittence. Because nonstrictly aquatic organisms implicitly have greater dispersal capacities, speciation rates would be lower and immigration rates greater than for strictly aquatic organisms at similar levels of intermittence, resulting in a lower slope of the decreasing richness pattern with flow intermittence. Blue and orange lines in the network represent perennial and intermittent river sections, respectively.

(Fig. 4.10.11). For instance, low levels of intermittence in headwaters should enhance speciation processes by long-term hydrological stability and isolation-by-distance mechanisms. As the intensity of intermittence increases, immigration should increase. Empirical evidence suggests that there is a higher number of aerial dispersal organisms and/or the role of overland dispersal in shaping aquatic communities increases in importance during the isolated-pools phase (Bonada et al., 2007; Razeng et al., 2016). Finally, with increasing flow intermittence, extinction processes should be more important and immigration should decrease as populations cannot persist, leaving no time for genetic differentiation and speciation. Of course, the importance of these processes could vary, resulting in different diversity patterns, if drying occurs in the lower reaches of the network or across the whole network.

Indirect evidence of differential evolutionary mechanisms in IRES shaping large-scale diversity patterns comes from global analyses of freshwater diversity. For instance, rivers located in Mediterranean and arid regions, where IRES are abundant, have high levels of endemism and beta-diversity (Leprieur et al., 2011). Macroinvertebrates, fish, and riparian communities of Mediterranean rivers harbor greater regional diversity, beta diversity, rarity, and endemism levels than temperate rivers (Bonada and Resh, 2013). Furthermore, Tedesco et al. (2012) suggest that the evolutionary processes shaping endemism in Mediterranean, semiarid, and arid regions differ from the mechanisms shaping the typically high levels of endemism found in tropical drainages. The diversity found at Mediterranean and arid latitudes may result from anagenetic speciation (i.e., the gradual transformation of a single species from ancestral species) through isolation and species range contractions. The intermediate position of these climatic regions between temperate and tropical climates makes them more sensitive to cycles of expansion and contraction at both longer and shorter geological time scales. For instance, global biome reconstruction for the Middle Pliocene (c. 3.6–2.6 Ma) suggests that tropical savannahs and woodland expanded in Africa and Australia at the expense of deserts (Salzmann et al., 2008), and pollen data suggests that the mid-Holocene (6000 years BP) climate was generally wetter than today in southern Europe and northern Africa (Wu et al., 2007). These historical climatic changes have certainly enhanced range expansions followed by contractions and isolation of previously connected populations, producing high levels of endemism and beta-diversity in IRES through local extinctions and speciation.

GLOBAL TRAIT PATTERNS IN IRES

IRES exhibit complex spatiotemporal flow patterns (Datry et al., 2016a; Chapters 2.2 and 2.3). Because differences in drying duration, predictability, and frequency among regions and within catchments have not yet been compiled globally, it is difficult to outline the “big picture” of how their aquatic assemblages have formed and which trait adaptations and strategies have been shaped and selected. Even if continental scale classifications of hydrological regimes exist (Chapter 2.3), the use of different methods impedes intercontinental comparisons of taxonomic and trait diversity from IRES. Although not all latitudes were included, Datry et al. (2014) adopted a global approach and compared 14 rivers from multiple biogeographic regions spanning a wide range of flow intermittence and spatial arrangements of perennial and IRES reaches. They found widespread congruence in the responses of invertebrate communities to intermittence in terms of richness and persistence strategies (resistance being less common than resilience). However, there was little or no influence of the spatial arrangement of perennial and IRES reaches, suggesting a common response of organisms regardless of the biogeographic species pool.

Resilience and resistance strategies are also common attributes found among biota from Mediterranean climate regions, with resilience being found more frequently than resistance (Hershkovitz and Gasith, 2013). Functional similarities (i.e., adaptive biological traits) among these regions have been widely recognized for a large variety of individual species and biological traits (Bonada and Resh, 2013), as have the dynamics of recovery following disturbance (a measure of the resilience of these ecosystems, Carmel and Flather, 2004). This overall functional convergence of individual species suggests that ecological and evolutionary processes driving patterns of trait adaptations to drying in IRES are deterministic and influenced by large-scale climatic and geomorphological factors shaping IRES globally, rather than being dependent on historical contingencies.

For a comprehensive evaluation of trait adaptations to intermittence, a wider comparison must include IRES from all climatic regions, spanning varying levels of magnitude, frequency, seasonal timing, predictability, and duration of dry periods. A recent analysis of datasets from temperate, Mediterranean, and arid regions examining the responses of aquatic macroinvertebrate assemblages to drying in relation to traits of resistance and resilience found that resistance strategies appear more prevalent in arid climates (Leigh et al., 2016b). However, nearby perennial sites are probably important sources of taxa with resilience strategies that can colonize IRES when flow returns (Chapter 4.8).

4.10.6 IRES ARE IDEAL SETTINGS FOR THE STUDY OF ECOLOGY AND EVOLUTION: CONCLUSIONS AND FUTURE RESEARCH

Evolutionary processes have been largely neglected in studies of IRES, despite the fact that these systems provide an ideal setting for studying the link between ecology and evolution. We base this conclusion on six main points. First, alternating wet and dry phases exert strong selection pressures in IRES that result in a myriad of adaptations. Second, IRES impose continual trade-offs for individuals, including how to respond to approaching dry phases, but also challenges for species stability and how to respond to disturbances. Third, the recurrent extinction and recolonization processes that affect IRES allow us to study how nonneutral and neutral processes influence genetic patterns and, similarly, how environmental sorting and dispersal influence the composition of local communities. Fourth, the high spatial and temporal habitat variability of IRES may lead to rapid niche differentiation and thus species diversification. Fifth, the drying period may impose changes in the relative roles of local and regional factors controlling biological communities and the relative importance of speciation, extinction, and immigration. Sixth, habitat contraction might change the relative effects of different evolutionary processes acting at population and community levels, which may ultimately influence EFs.

We hope our chapter stimulates future research on how ecology and evolution interact in IRES. In particular, studies should seek to understand (1) how evolutionary processes change in space and time; (2) the adaptive significance of resistance and resilience traits common in IRES biota; (3) how speciation, migration, and extinction processes vary across IRES networks; and (4) how evolutionary processes influence ecosystem-level processes in IRES. Future studies should not constrain themselves to single levels of organization (Fig. 4.10.1), but should focus on how evolutionary processes acting at one level influence those at other levels. The application of more powerful molecular techniques that are now available for the study of evolutionary ecology will help to elucidate some of the open questions presented here (Pauls et al., 2014). A more comprehensive view of the relationship between ecology and evolution in IRES will help us to predict how populations, communities, and ecosystems will adapt to increasing flow intermittence resulting from global change and other anthropogenic causes.

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